LIGHTYEAR

MS79 Services and API journey packs

Status: **implemented and contract-tested; customer execution remains unproven**.

Delivered in PR #209.

Why it matters

A service workload can be described in a reviewed pack instead of requiring a new hand-written driver for each customer journey. A prospect can inspect the requests, services and controls that a target run might touch before providing runtime access.

What was delivered

CloudBank's 18 journeys now come from a declarative pack. Scenario names, execution order and compatibility wrappers derive from that pack. Existing adapters retain responsibility for HTTP, authentication, messaging, platform controls and restoration.

A validator rejects unsafe or unsupported declarations. The planner traverses reachable macros, branches, loops, polling, parallel calls and cleanup to compute the declared invocation footprint and conservative maximum call counts. It does not merely repeat a pack author's claimed footprint.

The source lane receives only a reader for retained captures. It has no HTTP, control, submit or replay capability. Even a GET request is a target invocation, because an endpoint's HTTP method does not guarantee that it has no side effects. The target lane receives the invocation capabilities explicitly.

Try and inspect it

```
PYTHONPATH=src python -m lightyear_workflow.service_pack validate spec/service-packs/order-
approval.pack.json
PYTHONPATH=src python -m lightyear_workflow.service_pack plan spec/service-packs/order-
approval.pack.json
PYTHONPATH=src python -m lightyear_workflow.service_pack plan
src/lightyear_data/packs/cloudbank.services.json
PYTHONPATH=src python -m unittest tests.test_service_pack -v
```

Validation and planning need no runtime credentials or network access. The service pack guide explains the declaration language, calculated footprint, limits and adapter boundary. The repository includes both a small order-approval example and the CloudBank pack.

Evidence and boundaries

Contract tests cover rejected declarations, computed footprints, read-only source capabilities, bounded execution and compatibility with the CloudBank driver. Those tests establish local behavior, not a newly executed customer workload or fresh CloudBank equivalence. Physical hosts, credential-provider resources, network limits and control internals still belong to the trusted adapter's configuration. Declaring a normalization does not grant approval to use it.

This extends the declarative invocation pattern from batch packs to services and APIs. It provides a reusable invocation description; customer adapters and observed behavior still require qualification.